

Testing and Confidence Intervals for Variances

Exercise 1. Consider the following hypotheses:

$$H_0 : \sigma^2 = 169 \quad \text{versus} \quad H_1 : \sigma^2 \neq 169$$

A sample variance of 98 is obtained from a random sample of 10 observations drawn from a normally distributed population. At 1% significance level, conduct a test for the population variance using the critical value approach.

Exercise 2. A random sample of 21 pages is used to estimate the population variance of the number of typographical errors in a book. The sample mean and sample standard deviation are calculated as 7.34 and 5.11, respectively. Assume that the population is normally distributed.

- Construct a 95% interval estimate of the population variance.
- Construct a 99% interval estimate of the population variance.
- Use your answers to discuss the impact of the confidence level on the width of the interval.

Exercise 3. A financial analyst examines the performance of two mutual funds and claims that the variances of the annual returns for the bond funds differ. To support his claim, he collects data on the annual returns (in percent) for the years 2001 through 2010. The analyst assumes that the annual returns for the two emerging market bond funds are normally distributed. Here are some relevant summary statistics.

$$n_1 = 10 \quad s_1^2 = 500 \quad n_2 = 10 \quad s_2^2 = 100$$

The null hypothesis is, $H_0 : \frac{\sigma_1^2}{\sigma_2^2} = 1$ and $H_1 : \frac{\sigma_1^2}{\sigma_2^2} \neq 1$ since $s_1^2 > s_2^2$ (ensures we only need to think about the right tail).

- What is the critical F value at the 10% significance level?
- At $\alpha = 0.10$, is the analyst's claim supported by the data using the critical value approach?

Exercise 4. Students of two sections of a History course took a common final examination. The course instructor examines the variance in scores between the two sections. He selects random samples of $n_1 = 11$ and $n_2 = 16$ with sample variances of $s_1^2 = 400$ and $s_2^2 = 200$, respectively. Assuming that the population distributions are normal, construct a 90% confidence interval for the ratio of the population variance.

Exercise 5. 2008-9 Schol exam Q1

A team of builders has surveyed buyers of their new homes for years. Consistently, only 41% of the 2000 buyers have indicated that they were "quite satisfied" or "very satisfied" with the construction quality of their homes. The builders have adopted a revised quality-inspection system to try to improve customer satisfaction. They have surveyed 104 buyers since then; these buyers seem representative, with no systematic changes from past purchasers. Of the 104 buyers, 52 indicated they were quite or very satisfied.

- Formulate the null hypothesis that there has been no real change in customer satisfaction from the past rate. Explain the difference between the null and the alternative hypothesis.
- Explain the distinction between one-sided and two-sided alternatives. Which would you use here and why?
- Calculate the appropriate statistic for testing the null hypothesis.
- Test the null hypothesis at $\alpha = 0.05$.
- Construct a 95% confidence interval for the population difference.

Exercise 6. A researcher is interested in whether countries with floating exchange rates grow faster than those with fixed exchange rates. She gathers data from 1970-2009 for a number of countries and places each country-year pair in one of two groups depending on whether the country had a floating exchange rate in that year. This means some years a country might be in the group with floating exchange rates and other years in the group with fixed exchange rates. The sample mean annual growth rate for the 356 country-year pairs with floating exchange rates was 0.951% with a standard deviation of 5.860%, and other 1132 country-year pairs have a mean growth rate of 1.564% with a standard deviation of 5.034%.

- Assuming the population variances are the same is there any evidence that countries with floating exchange rates grow more quickly than those without?
- Test whether the assumption of equal population variances is valid and, if necessary, redo part a assuming the population variances are different.
- Does this tell us whether countries should adopt floating exchange rates? Be sure to explain why or why not.